

1. You are given a stream of lowercase characters. After each insertion, print the first non-repeating character so far. If none exists, print #. Return the resulting string.

Example

Input: "aabc"

Output: "a#bb"

Test Case 1

Input:

"abac"

Output:

"aabb"

Test Case 2

Input:

"zz"

Output:

"z#"

2. Given a binary string, return the length of the longest substring where:

Number of 0s = Number of 1s

All 0s appear before 1s

Test Case 1

Input:

"00110011"

Output:

4

Test Case 2

Input:

"00011101"

Output:

6

3. Given an array and integer K, Find minimum swaps required to bring all elements $\leq K$ together in a contiguous block.

Test Case 1

Input:

arr = [2, 1, 5, 6, 3]

K = 3

Output:

1

Test Case 2

Input:

arr = [1, 12, 10, 3, 14, 10, 5]

K = 8

Output:

2

4. You are given two strings containing only X and Y. You can swap adjacent "XY" to "YX" any number of times. Determine if you can transform string A into string B.

Test Case 1

Input:

A = "XXY"

B = "YXX"

Output:

true

Test Case 2

Input:

A = "YXYX"

B = "YYXX"

Output:

false

5. Given a grid of size $m \times n$, determine the number of distinct paths from the top-left corner (0,0) to the bottom-right corner ($m-1, n-1$). At each step, one can either move down or right.

Input: $m = 3, n = 3$

Output: 6

Explanation: Let the given input 3×3 grid is filled as such:

A B C

D E F

G H I

The possible unique paths which exists to reach 'I' from 'A' following above conditions are as follows:

ABCFI, ABEHI, ADGHI, ADEFI, ADEHI, ABEFI.

Input: $m = 2, n = 3$

Output: 3

Explanation: Let the given input 2×3 grid is filled as such:

A B C

D E F

The possible unique paths which exists to reach 'F' from 'A' following above conditions are as follows:

ABCF, ADEF and ABEF.

Input: $m = 1, n = 4$

Output: 1

Explanation: Let the given input 1×4 grid is filled as such:

A B C D

The only possible unique path is ABCD.